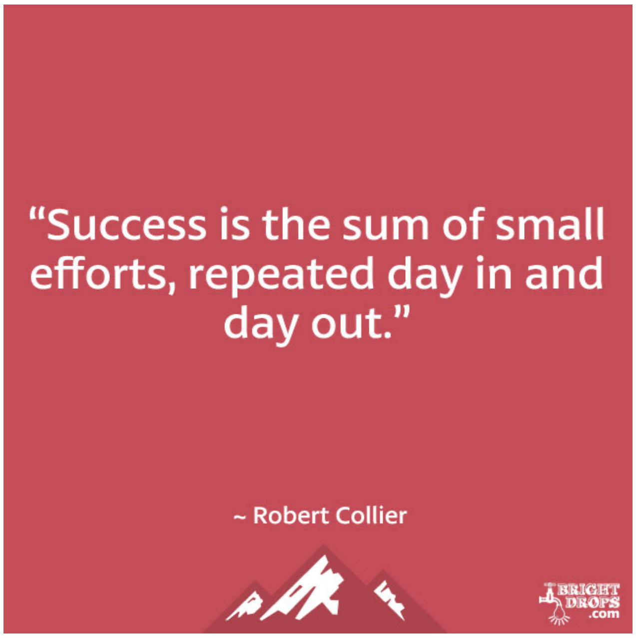
SUBARRAY



A very good Evening to Everyone :)

Today's Content

01. Subarray basics
02. Printing all subarrays
03. Printing all subarrays sums
04. Return total sum of all subarrays sums

↳ Google

Subarray

→ Continous part of an array is called a subarray

01. Single element is also a subarray
02. Whole array is also a subarray
03. Subarray is considered from left to right

ar[9] =	{ 4	1	2	3	-1	6	9	8	12 }
	0	1	2	3	4	5	6	7	8

Check if below are subarrays

{2 3 -1 6} : Yes

[2 5]

↑ indices

{4 2} : No

{1 2 6} No

{-1 6 9} : Yes

{8} : Yes

{8 9 6} : No

Properties of subarray

01. Given st & end of subarray.

length of subarray = end − st + 1

02. arr[ ] = {4 2 10 3 12 −2 15}
index: 0 1 2 3 4 5 6

Subarr starting from idx = 1

(1, 1) = {2}
(1, 2) = {2, 10}
(1, 3) = {2, 10, 3}
(1, 4) = {2, 10, 3, 12}
(1, 5) = {2, 10, 3, 12, −2}
(1, 6) = {2, 10, 3, 12, −2, 15}

Total = 6 subarrays

03. arr[ ] = {4 2 10 3 12 −2 15}
index: 0 1 2 3 4 5 6

Subarrays starting from idx 3

(3, 3) = {3}
(3, 4) = {3, 12}
(3, 5) = {3, 12, −2}
(3, 6) = {3, 12, −2, 15}

Ans = 4 subarrays

04. $arr[4] = \{\underset{0}{4} \quad \underset{1}{2} \quad \underset{2}{10} \quad \underset{3}{3}\}$

st = 0	st = 1	st = 2	st = 3
(0,0)	(1,1)	(2,2)	(3,3)
(0, 1)	(1,2)	(2, 3)	
(0, 2)	(1,3)		
(0, 3)			

Ans = 4+3+2+1 = 10

05 Generalise

$arr = \{a_0 \quad a_1 \quad a_2 \quad a_3 \; \ldots \; a_{n-2} \quad a_{n-1}\}$

stIdx = 0   stIdx = 1   . . .   stIdx = n−1

$$\underbrace{(0,0),\ (0,1),\ (0,2),\ (0,3),\ \ldots,\ (0,n-2),\ (0,n-1)}_{n} \qquad \underbrace{(1,1),\ (1,2),\ \ldots,\ (1,n-2),\ (1,n-1)}_{n-1} \qquad \underbrace{(n-1,n-1)}_{1}$$

Total subarr $= n + (n-1) + (n-2) + \cdots + 1$

= sum of n natural no.

$$\boxed{\text{Total subarray} = \frac{n(n+1)}{2}}$$

Q1. Given an arr[N], s, e, print subarray from [s e]

s → st point

e → end point

Idea → Iterate from s to e & print each & every ele

```
void print (int ar[n], int s, int e)
{
    for (i = s; i <= e; i++){
        print (ar[i]);
    }
}
```

TC = O(n)

SC = O(1)

Q2. Given ar[N], print all subarrays

Note :- Print each subarr in new line

Constraints :- $1 \le N \le 10^2$

$1 \le A[i] \le 10^5$

Eg :- ar[4] : { 2 8 -1 4 }

2	8	-1	4
0	1	2	3

$(0,0) = \{2\}$

$(0,1) = \{2, 8\}$

$(0,2) = \{2, 8, -1\}$

$(0,3) = \{2, 8, -1, 4\}$

$(1,1) = \{8\}$

$(1,2) = \{8, -1\}$

$(1,3) = \{8, -1, 4\}$

$(2,2) = \{-1\}$

$(2,3) = \{-1, 4\}$

$(3,3) = \{4\}$

Idea → To print a subarr, we need the st pt & end pt

$$\begin{array}{cccc} i\ \ j & j & j & j \\ (0,0) & (0,1) & (0,2) & (0,3) \end{array}$$

$$\begin{array}{ccc} i & & \\ (1,1) & (1,2) & (1,3) \end{array}$$

$$\begin{array}{cc} i & \\ (2,2) & (2,3) \end{array}$$

$$\begin{array}{c} i \\ (3,3) \end{array}$$

```
void printallsubarr ( int [] arr, int n )
{
    for ( i=0 ; i<n ; i++ ) {    // i = st point
        for ( j=i ; j<n ; j++ ) {    // j = end point
            st = i
            end = j
            for ( int k = st ; k <= end ; k++ ) {
                print (arr[k]);
            }
            System.out.println( );
        }
    }
}
```

Best

$TC = O(n^3)$

$SC = O(1)$

Total no. of subarrays $= \frac{n(n+1)}{2} * n$

$n \rightarrow$ print one subarray

To print all subarrays $= \frac{n^2(n+1)}{2}$

$= \frac{n^3 + n^2}{2}$

$= O(n^3)$

Q2. Given N arr elements, print each subarr sum

Note :- Print each subarr sum in new line

Constraints :- $1 \le N \le 10^3$

$1 \le ar[i] \le 10^5$

min sum

1

maxsum

$\underbrace{\{10^5, 10^5, 10^5, \ldots 10^5\}}_{10^3}$

$\rightarrow 10^5 * 10^3 = 10^8$

arr [4] =

6	8	-1	7
0	1	2	3

subarr	sum
[0 0]	= 6
[0 1]	= 14
[0 2]	= 13
[0 3]	= 20
[1 1]	= 8
[1 2]	= 7
[1 3]	= 14
[2 2]	= -1
[2 3]	= 6
[3 3]	= 7

Idea 1 → For every subarray, calculate the sum & print it

```
void printallsubarr ( int [] ar, int n )
    for ( i=0 ; i<n ; i++ ) {      // i = st point
        for ( j=i ; j<n ; j++ ) {  // j = end point
            st = i
            end = j        , sum = 0
            for ( int k = st ; k <= end ; k++ ) {
                sum += ar[k]
            }
            System.out.println(sum);
        }
    }
}
```

$TC = O(n^3)$

$SC = O(1)$

02. Use prefix array

int [] psum = new int [n];

```
void prefsum ( int [] arr, int n)
    01. Construct psum array → {TODO}

    for ( i= 0; i<n; i++){
        for ( j= i ; j<n ; j++){        L  R
                                        [i j]
            if (i==0) println (pf [j]);
            else { println ( pf [j] - pf [i-1]);
        }
    }
}
```

$TC = O(n^2)$

$\underbrace{SC = O(n)}$

$SC = O(1)$

03. Use carry forward approach

① → Given ar[N], print all subarrays sums starting at index 3.

ar[10] =

3	8	4	7	9	4	3	2	10	6
0	1	2	3	4	5	6	7	8	9

sum=0

subarrays	sum = 0
(3 3)	7
(3, 4)	
(3, 5)	
(3, 6)	
(3, 7)	
(3, 8)	
(3, 9)	

sum=0+7

sum=7+9

sum=16+4

sum=20+3

sum=23+2

sum=25+10

sum=35+6

```
void printsum3 (int [n] ar) {
    int sum= 0
    for (j=3 ; j<n; j++) {
        sum= sum+ ar[j] ;
        print(sum);
    }
}
```

TC= O(n)

SC= O(1)

Back to Original Ques

Idea 3 → Carry forward

```
for ( i=0 ; i<n; i++) {
      int sum= 0

      for (j = i ; j<n; j++) {
          sum= sum+ arr[j] ;
          print(sum);
      }
}
```

TC= $O(n^2)$

SC= $O(1)$

i= 0   j [0 n-1] → print all subarray st from 0

i = 1   j [1 n-1] → print all subarray st from 1

i= 2   j [2 n-1] → ,, ,, ,, ,, ,, 2

⋮

i=n-1   j [n-1 n-1] → ,, ,, ,, ,, ,, n-1

Q → Given an arr[n], return sum of all subarr sum.

arr[4] = { 6, 8, -1, 7 } } Ans = 94

(indices: 0, 1, 2, 3)

Subarrays		sum
(0,0)	=	6
(0,1)	=	14
(0,2)	=	13
(0,3)	=	20
(1,1)	=	8
(1,2)	=	7
(1,3)	=	14
(2,2)	=	-1
(2,3)	=	6
(3,3)	=	7
		94

Brute force → For every subarr, get the sum & add it to the total sum

Approach 1

3 nested loops

TC = $O(n^3)$

SC = $O(1)$

Approach 2

Prefix array

TC = $O(n^2)$

SC = $O(n)$

Approach 3

Carry forward

TC = $O(n^2)$

SC = $O(1)$

```
int  totalsum = 0

for( i=0 ; i<n; i++){
    int  sum= 0
    for( j= i ; j<n; j++){
        sum= sum+ ar[j];
        totalsum+= sum;
    }
}
return totalsum
```

$TC = O(n^2)$

$SC = O(1)$

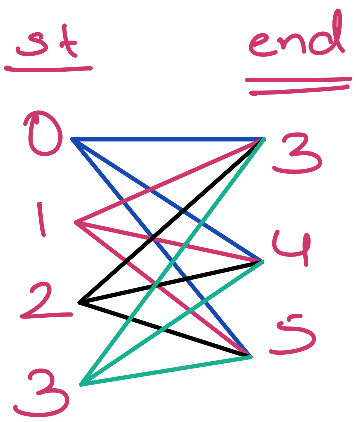
$\underline{\underline{TC = O(n)}}$ $\longrightarrow$ Constraint

Final Idea { Contribution technique }

ar[6] =

3	-2	4	-1	2	6
0	1	2	3	4	5

In how many subarrays, index 3 is present?



(0,3) (1,3) (2,3) (3,3)

(0,4) (1,4) (2,4) (3,4)

(0,5) (1,5) (2,5) (3,5)

Total no. of subarrays containing index 3 $= 4 * 3 = 12$

$arr[6] =$

3	-2	4	-1	2	6
0	1	2	3	4	5

In how many subarrays, index 1 is present?

st	end
0	1
1	2
	3
	4
	5

Total no. of subarrays $= 2 * 5 = 10$

Q Given N elements, how many subarrays will contain i

$$arr = \{ \underbrace{a_0 \; a_1 \; a_2 \; a_3 \; .. \; a_{i-1} \; a_i}_{st[0 \;\; i]} \; a_{i+1} \; ... \; a_{n-1} \}$$

$st[0 \;\; i]$ $\Downarrow$ $\#(i+1)$

$end[i \;\; n-1]$ $\Downarrow$ $\#(n-i)$

$$Ans = (i+1) * (n-i)$$

$$
\begin{array}{llcccc}
 & & 0 & 1 & 2 & 3 \\
arr[\,] & = & \{\,6 & 8 & -1 & 7\,\} \\
 & & & & & \\
(i+1) & = & 1 & 2 & 3 & 4 \\
* & & * & * & * & * \\
(n-i) & = & 4 & 3 & 2 & 1 \\
\hline
\text{Total} & = & \boxed{4} & \boxed{6} & \boxed{6} & \boxed{4}
\end{array}
$$

In 4 subarr 6 will be present

In 6 subarr, 8 will be present

In 4 subarr 7 will be present

arr = {6, 8, −1, 7}

Subarrays

(0,0) = {6} = 6

(0,1) = {6, 8} = 6 + 8

(0,2) = {6, 8, −1} = 6 + 8 + (−1)

(0,3) = {6, 8, −1, 7} = 6 + . . .

(1,1) = {8}

(1,2) = {8, −1}

(1,3) = {8, −1, 7}

(2,2) = {−1}

(2,3) = {−1, 7}

Total contribution of an ele in all subarrays

$6 * 4 = 24$

$8 * 6 = 48$

$-1 * 6 = -6$

$7 * 4 = 28$

$\overline{\;94\;}$

$(3,3) = \{7\}$

94

```
int tot = 0
for( i=0; i<n; i++){
    int nt = (i+1)*(n-i);
    tot += arr[i] * nt;
}
```

TC = O(n)

SC = O(1)

$0 \le len(A) \le 7e^5 \Rightarrow 0 \le len(A) \le 10^5$

$1 \le A[i] \le 1e^7 = 1 \le A[i] \le 10^7$